

10.2 Factorising a difference of perfect squares



A common and key step in the simplification and solution of equations involves factorisation. Factorisation is the process of writing a number or expression as a product of its factors. For example: $6 = 2 \times 3$, $2x + 6 = 2(x + 3)$, $x^2 - x = x(x - 1)$ and $x^2 - 9 = (x + 3)(x - 3)$.

In this section we look at expressions in which each term has a common factor and expressions that are a difference of perfect squares.

► Let's start: It's just a DOPS expansion in reverse

Complete each column to see the connection when expanding or factorising a DOPS.

Expand

$$(x + 2)(x - 2) = x^2 - 4$$

$$(x - 3)(x + 3) = x^2 - 9$$

$$(2x + 3)(2x - 3) = 4x^2 - 9$$

$$(7x - 6)(7x + 6) = \underline{\hspace{2cm}}$$

Factorise

$$x^2 - 4 = (x \underline{\hspace{1cm}})(x \underline{\hspace{1cm}})$$

$$x^2 - 9 = (x \underline{\hspace{1cm}})(\underline{\hspace{1cm}})$$

$$4x^2 - 9 = (2x \underline{\hspace{1cm}})(\underline{\hspace{1cm}})$$

$$\underline{\hspace{2cm}} = (\underline{\hspace{1cm}})(\underline{\hspace{1cm}})$$

Key ideas

- **Factorise** expressions with common factors by 'taking out' the highest common factor.

For example: $-5x - 20 = -5(x + 4)$ and $4x^2 - 8x = 4x(x - 2)$

- Factorise a difference of perfect squares (DOPS) using

$$a^2 - b^2 = (a + b)(a - b)$$

For example:

$$\begin{aligned} x^2 - 16 &= x^2 - 4^2 \\ &= (x + 4)(x - 4) \end{aligned}$$

and

$$\begin{aligned} 9x^2 - 25 &= (3x)^2 - 5^2 \\ &= (3x + 5)(3x - 5) \end{aligned}$$

Factorise To write an expression as a product, often involving brackets

Exercise 10B

- 1 Complete these statements.

- a** $2(x + 3) = 2x + 6$ so $2x + 6 = 2(\underline{\hspace{1cm}})$
b $-4(x - 1) = -4x + 4$ so $-4x + 4 = -4(\underline{\hspace{1cm}})$
c $(x + 2)(x - 2) = x^2 - 4$ so $x^2 - 4 = (\underline{\hspace{1cm}})(\underline{\hspace{1cm}})$
d $(3x + 2)(3x - 2) = 9x^2 - 4$ so $9x^2 - 4 = (\underline{\hspace{1cm}})(\underline{\hspace{1cm}})$

- 2 Determine the highest common factor of these pairs of terms.

- a** $7x$ and 14 **b** $12x$ and 30 **c** $-8y$ and 40
d $-5y$ and -25 **e** $4a^2$ and $2a$ **f** $12a^2$ and $9a$
g $-5a^2$ and $-50a$ **h** $-3x^2y$ and $-6xy$

Understanding

Include a common negative.



Fluency

Example 5 Taking out common factors

Factorise by taking out the highest common factor.

a $-3x - 12$

b $20a^2 + 30a$

Solution

a $-3x - 12 = -3(x + 4)$

b $20a^2 + 30a = 10a(2a + 3)$

Explanation -3 is common to both $-3x$ and -12 .The HCF of $20a^2$ and $30a$ is $10a$.**3** Factorise by taking out the highest common factor.

a $3x - 18$

b $4x + 20$

c $7a + 7b$

d $9a - 15$

e $-5x - 30$

f $-4y - 2$

g $-12a - 3$

h $-2ab - bc$

i $4x^2 + x$

j $5x^2 - 2x$

k $6b^2 - 18b$

l $14a^2 - 21a$

m $10a - 5a^2$

n $12x - 30x^2$

o $-2x - x^2$

p $-4y - 8y^2$

Find the highest common factor then take it out.

**Example 6** Factorising a difference of perfect squares

Factorise the following differences of perfect squares.

a $x^2 - 16$

b $9a^2 - 4b^2$

Solution

a $x^2 - 16 = (x)^2 - (4)^2$
 $= (x - 4)(x + 4)$

b $9a^2 - 4b^2 = (3a)^2 - (2b)^2$
 $= (3a - 2b)(3a + 2b)$

ExplanationUse $a^2 - b^2 = (a - b)(a + b)$ with $a = x$ and $b = 4$. $9a^2 = (3a)^2$ and $4b^2 = (2b)^2$.**4** Factorise the following differences of perfect squares.

a $x^2 - 9$

b $x^2 - 25$

c $y^2 - 49$

d $y^2 - 1$

e $a^2 - 16$

f $b^2 - 36$

g $y^2 - 144$

h $z^2 - 400$

i $4x^2 - 9$

j $36a^2 - 25$

k $1 - 81y^2$

l $100 - 9x^2$

m $25x^2 - 4y^2$

n $64x^2 - 25y^2$

o $9a^2 - 49b^2$

p $144a^2 - 49b^2$

 $a^2 - b^2 = (a + b)(a - b)$ **5** Factorise these differences of perfect squares.

a $4 - x^2$

b $9 - y^2$

c $36 - a^2$

d $100 - 9x^2$

e $b^2 - a^2$

f $400 - 25a^2$

g $4a^2 - 9b^2$

h $16y^2 - 121x^2$

Problem-solving and Reasoning

Example 7 Factorising by first taking out a common factorFactorise $12y^2 - 1452$ by first taking out a common factor.**Solution**

$12y^2 - 1452 = 12(y^2 - 121)$
 $= 12(y - 11)(y + 11)$

Explanation

First take out the common factor of 12.

 $121 = (11)^2$, use $a^2 - b^2 = (a - b)(a + b)$.

- 6 Factorise the following by first taking out a common factor.

a $2x^2 - 32$ **b** $5x^2 - 45$ **c** $6y^2 - 24$
d $3y^2 - 48$ **e** $3x^2 - 75y^2$ **f** $3a^2 - 300b^2$
g $12x^2 - 27y^2$ **h** $63a^2 - 112b^2$ **i** $108x^2 - 147y^2$

As a first step, take out a common factor then factorise the DOPS.



- 7 The height (in metres) of a falling object above ground level is given by $100 - t^2$, where t is in seconds.

- a** Find the height of the object:
 i initially (at $t = 0$)
 ii after 2 seconds
 iii after 8 seconds
b Factorise the expression $100 - t^2$.
c Use your factorised expression from part **b** to find the height of the object:
 i initially (at $t = 0$)
 ii after 2 seconds
 iii after 8 seconds
d How long does it take for the object to hit the ground?



- 8 We can work out problems such as $19^2 - 17^2$ without a calculator like this:

$$\begin{aligned}
 19^2 - 17^2 &= (19 + 17)(19 - 17) \\
 &= 36 \times 2 \\
 &= 72
 \end{aligned}$$

Factorise first, using $a^2 - b^2 = (a + b)(a - b)$, then evaluate.

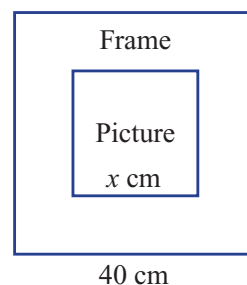


Use this idea to evaluate the following by first factorising, without the use of a calculator.

a $16^2 - 14^2$ **b** $18^2 - 17^2$ **c** $13^2 - 10^2$ **d** $15^2 - 11^2$
e $17^2 - 15^2$ **f** $11^2 - 9^2$ **g** $27^2 - 24^2$ **h** $52^2 - 38^2$

★ Flexible framing

- 9 A special picture frame can hold a square picture of any size up to 40 cm.
- a** Using a picture side length of x cm, write expressions for the area of:
 i the picture **ii** the frame (not including the picture)
b Factorise your expression for the frame area.
c Find the frame area if: **i** $x = 20$ **ii** $x = 10$
d Using trial and error, what value of x is required if the frame area is to be 700 cm^2 ?



Using technology 10.2: Expanding and factorising

This activity is available on the companion website as a printable PDF.